

P2 Wedge — Simulation Campaign

For Alexandra · prepared by Dylan (+ Claude) ·
2026-08-05 · works offline once downloaded

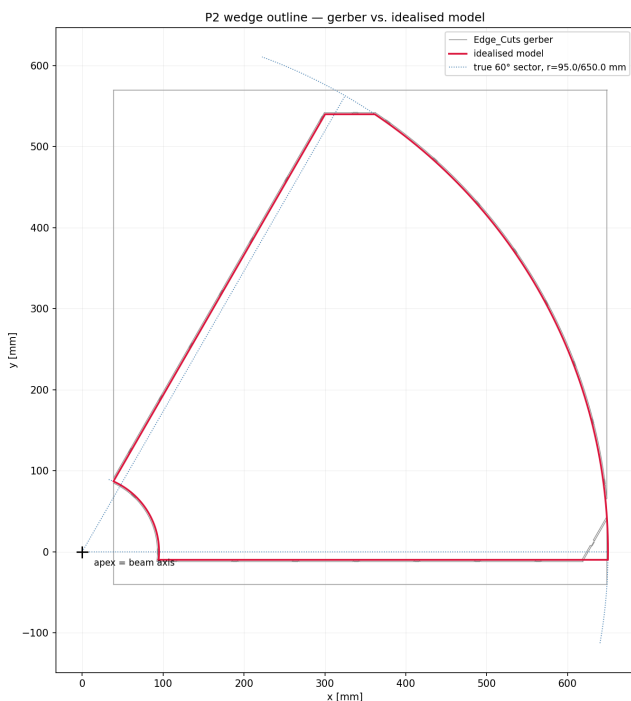
This document is everything we have so far on the Geant4 simulation of the P2 wedge Micromegas: the detector as we modelled it, the toolchain we plan to use, our understanding of the experiment, and the full campaign plan. It is also a **review request**: every number we had to guess is marked **GUESS**, confirmed ones **CONFIRMED**, and the things only you can answer are collected in the purple boxes. Nothing needs answering today — read at leisure, annotate mentally, and we'll go through it when you land.

- 1 · The detector as modelled
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1 · The detector as modelled

The model lives in a new `p2` mode of the (MX17-derived) Geant4 app, with a Python mirror used for all figures here. Coordinates are the **gerber frame**: the wedge apex — the MESA beam axis — is at (0,0), so the pad-mapping files and simulation output share one frame with no offsets. $z = 0$ at the front window plane, $+z$ along the beam.



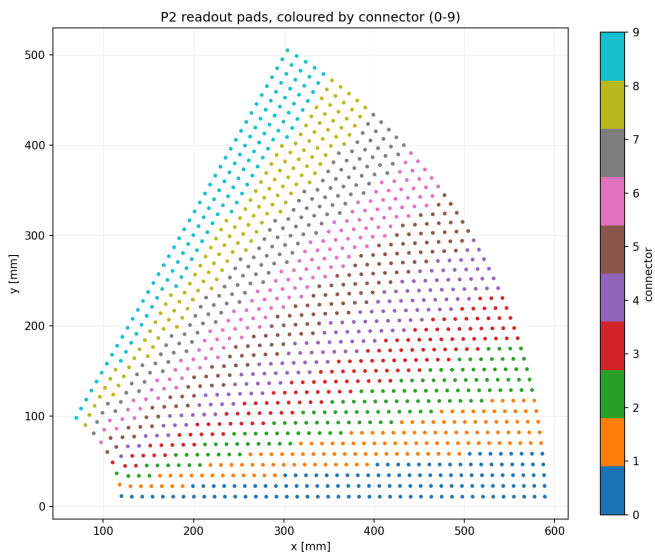
Board outline from the Edge_Cuts gerber (red) vs the idealised model: 60° annular sector, board $r = 95\text{--}650$ mm, radial edges offset +10 mm, top cut at $y = 540$ mm. The only unmodelled feature is a small chamfer bottom-right.

Active area	$r = 120 \rightarrow 590$ mm, $\phi = 0.97^\circ \rightarrow 59.0^\circ$, ≈ 1689 cm ²
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Readout	1280 pads = 10 connectors $\times$ 128 ch; 42 rings, 11.43 mm radial pitch, area-equalized ($\approx 11.43 \times 11.86$ mm cells)
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Pillars	$\varnothing$ 0.5 mm on an exactly 2.000 mm pitch $\rightarrow$ 41 366 pillars, 4.8 % of the amp-gap area
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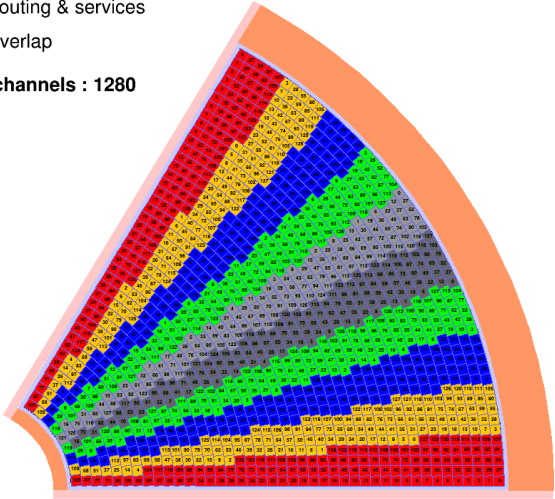
Readout PCB	2 layers, 0.236 mm total: 18 μm Cu / 200 μm FR4 / 18 μm Cu (Stack_Up_P2.txt)
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The 1280-pad readout map from the channel-mapping files: 42 rings with 9–51 pads each. Not a regular grid — pad lookup is ring-then-azimuth.

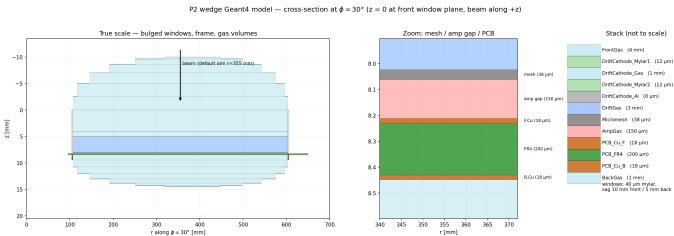
- Nominal module : $110 < R < 600$ mm ; 60 deg.
- Active area : 95.3 %
- Routing & services
- Overlap

Nb. of channels : 1280



The area-equalization of the pad cells across the wedge (from design/mapping).

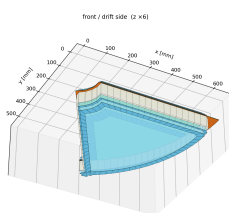
The stack, front to back



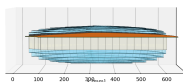
Cross-section of the modelled stack at $\phi = 30^\circ$ (left: true scale with bulged windows; middle: zoom on mesh/amp gap/PCB; right: layer key). Text is small — pinch to zoom, or use the table below.

z [mm]	layer	as modelled	status
-10 → 0	front window	40 μm mylar, bulged, 10 mm sag	GUESS thickness & sag
0 → 4.0	front gas	4 mm chamber gas	from frame STEP
4.0	cathode foil 1	12 μm mylar	two-foil CONFIRMED , thickness GUESS
4.01 → 5.01	inter-foil gas	1 mm ("maybe 1 mm")	GUESS
5.01	cathode foil 2	12 μm mylar + 0.1 μm Al facing drift	Al side CONFIRMED , thickness GUESS
5.02 → 8.02	drift gas (sensitive)	3.0 mm baseline, scans 1-4 mm	CONFIRMED
8.02	micromesh	woven SS 48 μm opening / 19 μm wire → 38 μm slab at 22.3 % density	spec CONFIRMED ("48×19")

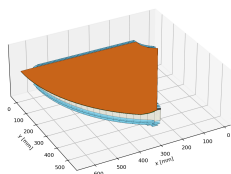
8.06 → 8.21	amp gas (sensitive)	150 μm	CONFIRMED
8.21 → 8.45	readout PCB	18 μm Cu / 200 μm FR4 / 18 μm Cu	exact (fab file)
8.45 → 9.45	back gas	1 mm = carbon back- frame depth	range CONFIRMED (1 mm normal, ≤ 3 mm this production)
9.45 →	back window	40 μm mylar, bulged, 5 mm sag, glued to carbon frame	anatomy CONFIRMED , numbers GUESS



P2 wedge Geant4 model — 3D (z exaggerated for visibility)
edge-on — window bulge (z x6)

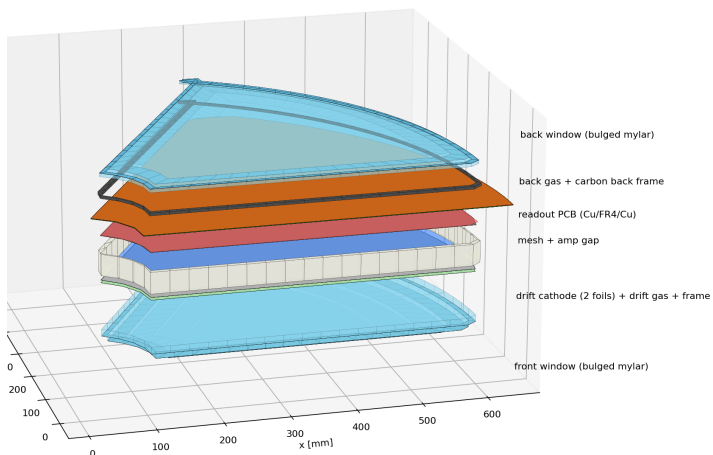


back / readout side (z x6)



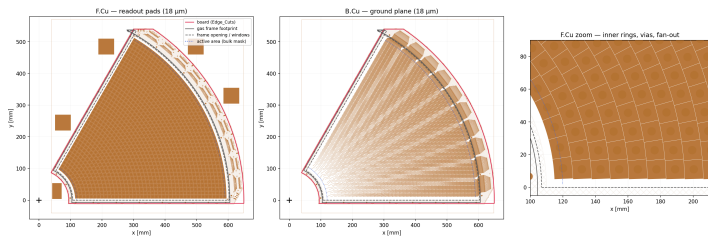
3D view of the assembled model.

P2 wedge model — exploded (z x4, groups separated)



Exploded view: windows, cathode foils, mesh, PCB, frames.

P2 readout board — production gerbers (P2_BASKET_Apr26) with model overlays

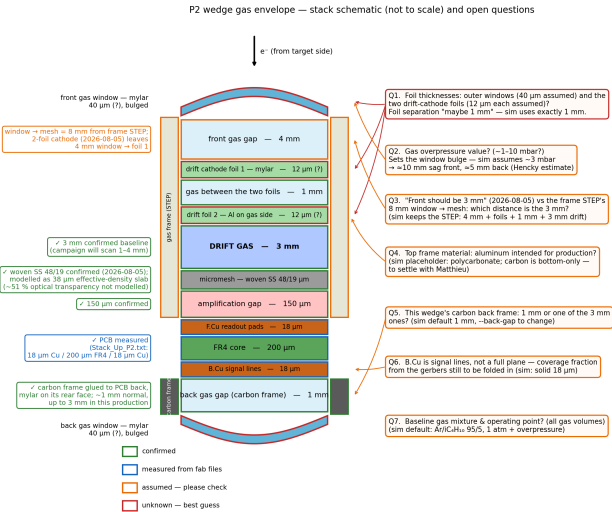


The real copper layout of the board (front pads / back signal lines). Note the back copper is signal lines, not a full plane — the model still uses a solid 18 μm sheet (conservative).

Why the windows bulge in the model

The chamber runs a few mbar above atmosphere, so the two outer mylar sheets bloat outward. We model each window as a *terraced dome* — six stacked gas prisms shrinking by a spherical-cap law, each capped by a flat mylar annulus, so every straight-through path crosses exactly one 40 μm mylar thickness. The sag height comes from Hencky's membrane formula: at ~ 3 mbar with a ~ 220 mm half-span it gives ≈ 10 mm; because sag scales as $\text{pressure}^{1/3}$, the whole plausible 1–10 mbar range only spans 7–15 mm, so this is not a sensitive parameter.

2 · Questions on the geometry



One-page summary of the open geometry questions (same list as below).

For you, roughly by physics impact:

1. **Foil thicknesses** — outer windows (we assumed 40 μm) and the two drift-cathode foils (we assumed 12 μm each, 1 mm apart). Actual values?
2. **“The front should be 3 mm”** — your earlier comment vs the frame STEP's 8 mm window→mesh height. Which is right, or what does “front” refer to?
3. **Top frame material** — is aluminum the production intent (we heard it's under discussion, carbon never on top)? Currently polycarbonate placeholder. Push to Matthieu if needed.
4. **This wedge's back frame** — 1 mm, or one of the ≤ 3 mm ones from this production?
5. **Overpressure** — a mbar value (or a measured window sag) would pin the bulge model.
6. **Back copper** — we'll extract the real signal-line coverage from the gerbers; anything to know about ground fills?
7. **Mesh reading** — we read “48×19 opening and pitch” as 48 μm opening / 19 μm wire (67 μm pitch, ≈ 51 % optical transparency). Correct?

8. **Pad mapping revision** — the two mapping file sets disagree on 79/1280 channels and use opposite radial ordering. Which one does the DAQ use?

3 · The toolchain, stage by stage

We deliberately do *not* run one monolithic Geant4+Garfield binary. The campaign uses the staged pattern (as in ATLAS NSW digitization and the CAST/IAXO chains): three stages joined by files, each re-runnable alone, so a change in — say — the electronics threshold never re-runs particle transport.

Stage A — Geant4 (this repo)

full geometry · EM transport · PAI in gas

every material, photon conversion, δ -ray;
fluorescence/Auger on (Livermore-class EM)

*ClusterTree: ionization clust
+ track segments for Heed*

Stage B — detector response (Python)

Magboltz gas tables · Garfield++/Heed anchors

drift + diffusion $\sigma\sqrt{z}$ · mesh transparency ·
Polya-fluctuated gain (Penning-corrected) ·
charge summed onto the 1280 real pads

pad hits: (pad, charge [fC], ...)

Stage C — electronics + analysis

VMM3a-informed threshold/ADC model

ENC noise floor at 75–170 pF · neighbor logic ·
ADC saturation · then: efficiency, ROC curves,
pad multiplicity, drift-time span

Stage A — Geant4

Owens everything that needs the full geometry: transport of the beam electrons through every layer, photon conversions in the solids, δ -rays, fluorescence. Physics choices that matter here: Livermore-class EM with atomic deexcitation on (so Cu-K 8.0 keV and Fe-K 6.4 keV fluorescence from

pads and mesh are produced), and the **PAI model** in a region over the gas volumes — plain condensed-history EM gets the *mean* ionization right in 3 mm of gas but not the event-by-event fluctuations, which is exactly what efficiency depends on. Output per event: ionization clusters (position, energy, primary-electron count, provenance) plus the track segments crossing the drift gas.

Stage B — gas response (Magboltz + Garfield++/Heed, then a fast sampler)

Magboltz is run once per gas mixture to tabulate drift velocity, longitudinal/transverse diffusion, Townsend and attachment coefficients on a field grid spanning drift (~ 100 V/cm–1 kV/cm) and amplification (~ 30 –60 kV/cm) fields; the tables are cached and never regenerated per job. **Penning transfer must be switched on** — without it simulated Micromegas gains come out low by large factors. Published transfer rates exist for Ar/Iso and Ne/CO₂ but *not* for Ne/Iso or any

ternary, so for those we set r manually and bracket it — one of the honest systematic uncertainties.

Garfield++ is used surgically: Heed re-ionizes the exported track segments when true cluster statistics matter, and AvalancheMicroscopic runs once per (gas, HV) point on $\sim 10^3$ – 10^4 single electrons to extract mean gain, Polya shape parameter and mesh transparency. Those numbers are then frozen into a fast Python sampler — Gaussian diffusion $\sigma/\sqrt{z}$, mesh transparency, Polya-fluctuated gain, charge summed onto the real 1280-pad map — which costs microseconds per electron, so the full campaign is cheap to re-run.

Stage C — VMM3a electronics

The front-end is VMM3a (64 ch, shaper peaking 25–200 ns, gains 0.5–16 mV/fC, 10-bit peak ADC, neighbor logic). Baseline model: charge-integration per pad + threshold — at 200 ns peaking essentially the whole Micromegas signal is integrated — with three corrections we consider mandatory: (i) the threshold expressed in

ENC units at the real pad capacitance (75–170 pF → several-thousand-electron noise, few-fC practical thresholds; your expected signal ~ 21 fC); (ii) **neighbor logic**, without which cluster size and charge are underestimated — and that directly biases the electron-vs-photon topology cut; (iii) **ADC saturation** — at high gain a 6–8 keV photoabsorption (~ 250 electrons at a point) can saturate the ~ 8 -bit range while MIP pads stay in range. Full waveform simulation (ATLAS Athena VMM shaper $h(t)$) is held in reserve for timing/pile-up studies.

4 · The experiment as we understand it

Please sanity-check this section — it mixes public sources with our own inference.

P2 @ MESA: 155 MeV, 150 μ A polarized electrons on a 600 mm LH₂ target; parity-violating elastic asymmetry → $\sin^2\theta_W$ to ~ 0.14 %. **BASKET** adds backward angles ($\theta = 140^\circ$ – 150° , full azimuth): three wheels $\times$ 6 wedges $\times$ 1280 pads $\approx 23\,000$ channels,

trigger-less VMM3a streaming, measuring G_M^s and G_A to cut the hadronic uncertainty of the forward measurement $\sim 4\times$.

	value	our confidence
Electron energy at the wedges	119 MeV (155 beam, elastic at $140\text{--}150^\circ$); 144 MeV for the 200 MeV option; radiative tail below	computed, solid
Incidence angle	$\sim 10^\circ$ from wedge normal <i>if</i> wheels sit in the fringe field ($B \approx 50$ mT); would be $30\text{--}40^\circ$ inside the bore	ASSUMPTION — scan $0\text{--}40^\circ$
B-field at the wedges	$\approx$ none (wheels outside the 0.6 T solenoid) $\rightarrow$ straight tracks, no field in the sim	ASSUMPTION
Rates	~ 100 MHz backward electrons total, $15\text{--}20$ kHz/pad; photon background $50\text{--}150$ keV at $\sim 10^3\times$ the electron rate	public numbers

Experiment questions for you:

1. Wheel z-positions and the local B-field — is the $\sim 10^\circ$ incidence picture right? A field map would settle it.
2. Is $50\text{--}150\text{ keV} \times \sim 10^3$ the right photon background band and ratio, and is there a photon-shield design we should include eventually?
3. What gas does the collaboration currently baseline, and has neon been discussed before?

5 · The campaign: goals, phases, parameters

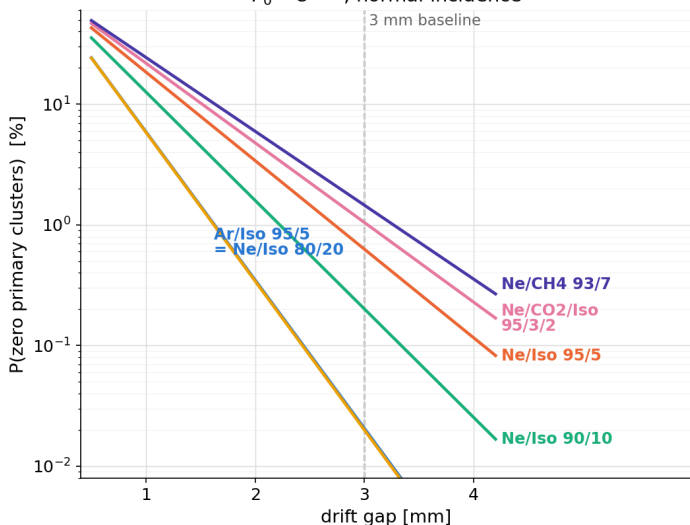
The single question underneath everything: **how well can one wedge detect 30–160 MeV electrons while rejecting 50–150 keV photons arriving at a thousand times the rate** — and how does that trade off against drift gap and gas choice? Concretely, per (gas, gap) configuration we want: electron efficiency above threshold; photon fake-hit probability at the same threshold, broken down by *which layer*

converted the photon (a low-Z gas only helps against gas conversions — wall conversions set a floor); the ROC of the two as the charge cut scans; and the operational numbers (pad multiplicity for streaming bandwidth, drift-time span for the coincidence window, required mesh voltage).

The neon idea. MX17 experience suggests neon mixtures are far less sensitive to the keV-range photons while costing only ~30 % of the electron signal. But neon's primary-cluster density is lower, and at small gaps that eats into efficiency — the plot below is the zeroth-order version of the whole trade-off. $P_0 = e^{(-n_p \cdot d)}$ is the probability a crossing electron leaves *zero* primary clusters: a hard inefficiency floor no gain or threshold can recover.

Hard inefficiency floor vs drift gap

$$P_0 = e^{-n_p d}, \text{ normal incidence}$$



The hard inefficiency floor: Ne/Iso 95/5 at 3 mm sits where Ar/Iso sits at ~1.7 mm. Quencher fraction is a strong knob (90/10, 80/20) — but flammability limits it (§6). Real efficiency after the ADC threshold is worse; computing it honestly is Phase 2's job.

Phases

phase	what	status
0	Code prep: gas-mixing bug fix (vol vs mass fractions — fatal for Ne mixes), new gases + W-values, gun angle control, cluster provenance, PAI region, first Ixplus build, Magboltz tables, Stage B skeleton, condor refresh	can start now, none of it waits on your answers

1	Geant4 sensitivity scans — electrons 1–200 MeV × 0–40°, photons 3–500 keV; deliverables: sensitivity curves per gas, photon conversion-layer budget	after your geometry review
2	Full response chain on baseline: pad charge maps, ROC curves (ϵ_e at 90/99/99.9 % photon rejection), coincidence-rate extrapolation	after 1
3	The decision matrix: 7 drift gaps (1–4 mm) × ~8 gases × ~9 kinematic points ≈ 500 runs on HTCondor → (gap, gas) heatmaps + recommendation memo	after 2
4	Optional: VMM waveforms, woven mesh, Degrad spot-checks, radiative-tail source spectra	if needed

Scan axes (Phase 1 grids)

Electron energy	1–200 MeV, 15 points, dense near 119/144 MeV
Photon energy	3–500 keV, 17 points (3–10 keV covers Ar-K/Fe-K/Cu-K structure; 50–150 keV = the background band)

Incidence angle	0, 10, 20, 30, 40°
Drift gap	1, 1.5, 2, 2.5, 3, 3.5, 4 mm
Position	mid-active default ($r = 355$ mm, $\phi = 30^\circ$); inner/outer/edge check at baseline
Statistics	10^4 – 10^5 electrons, 10^6 – 10^7 photons per point (empty events skipped on write)

6 · Gas candidates

Twelve candidates go through Magboltz + Phase 1; ~8 survive into the full matrix. Flammability is computed per ISO 10156:2017 — notably *all* ≥ 5 % isobutane mixes are flammable (the folklore “ <3 % iso is safe” traces to the superseded 1996 edition), and CO_2 is what rescues the ternaries. Isobutane's vapor pressure also forbids full-pressure premix bottles above ~1.5 %, so serious isobutane options imply on-site mixing (a standard 3-channel MFC rack, ~2–4 k€/channel).

#	mixture	flammable?	premix?	why it here
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A1	Ar/Iso 95/5	yes	no	current default referrer
A2	Ar/CO ₂ /Iso 93/5/2	no (barely)	on-site	ATLAS NSW heritage
A3	Ar/CF ₄ /Iso 88/10/2	no	no	timing referrer
N1	Ne/Iso 95/5	yes	no	the original proposal
N2	Ne/Iso 90/10	yes	no	quench recovery
N3	Ne/Iso 80/20	yes	no	full Ar- equivalent n_p
N4	Ne/CF ₄ 90/10	no	yes	fast option
N5	Ne/CO ₂ /Iso 95/3/2	no (margin)	on-site	best- margin ternary
N6	Ne/Iso/CO ₂ 90/5/5	no (PICOSEC)	on-site	PICOSEC eco-mi measured gain >
N7	Ne/CH ₄ 93/7	no	yes	simple non- flamm binary
N8	Ne/C ₂ H ₆ 90/10	yes	yes	COMP/4 chemical cans C

Honest caveats we carry through all results: assumed Penning r for Ne/Iso, Ne/CH₄ and ternaries; no published gain curve for *any* Ne mix in a 150 μm bulk (a ⁵⁵Fe/⁹⁰Sr scan on the prototype would be both needed calibration and publishable); mesh modelled as a solid slab (overestimates mesh photon conversions); pillars (4.8 % of amp gap) ignored; the “Ne runs at lower voltage” lore is unverified — the defensible claims are ~5× max-gain headroom and much lower discharge probability.

7 · The frameworks in depth — how each one actually works

Section 3 said *what* each stage does; this one explains *how the tools underneath actually work*, because knowing their mechanics is what tells you which one to trust for which question. (Full citations live in the repo,

`docs/research/TOOLCHAIN_NOTES.md`.)

Geant4 — a particle random walk through the geometry

Geant4 is a C++ toolkit in which you describe the detector as nested volumes of declared materials, and it then transports particles through them **step by step**. For each step it asks every physics process attached to the particle ("how far until you'd photo-absorb? Compton-scatter? ionize? bremsstrahlung?"), samples a distance from each cross-section, and the shortest one wins — the particle advances, the interaction happens, secondary particles go onto a stack and are tracked in turn. The catalogue of processes is the *physics list*; we use a Livermore-class electromagnetic list, which carries the low-energy atomic data we need, with atomic deexcitation switched on so that when a photon knocks a K-shell electron out of copper, the 8.0 keV Cu fluorescence photon (or Auger electron) that follows is actually produced and tracked.

One crucial approximation to understand: for charged particles Geant4 does

condensed-history transport — it does not simulate every one of the thousands of tiny collisions an electron makes per millimetre, it lumps them into one net energy loss and deflection per step, drawn from a fluctuation model. That is fine through a PCB; it is *not* fine in 3 mm of gas, where we care about the exact number of ionization clusters — the lumped model gets the mean right but the event-by-event spread wrong, and the rare zero-cluster events *are* our inefficiency. The fix at the Geant4 level is the **PAI (photo-absorption ionization) model**, enabled just in the gas regions: it replaces the generic fluctuation model with one built from the gas's actual photoabsorption spectrum, so thin-layer energy-loss straggling comes out right, at about 2× the CPU cost.

In our app, the drift and amplification gas volumes are declared sensitive: every energy-depositing step there is recorded as a cluster (position, energy, estimated electron count, and — after Phase 0 — the provenance: which process and which layer the depositing track was born in). That last

bit is what turns a photon run into a per-layer conversion budget.

Heed — ionization done properly, cluster by cluster

Heed (Smirnov) is a specialist: given a charged particle crossing a gas, it returns the actual **list of ionizing collisions** — where each cluster is, and how many electrons it contains — sampled from the same photoabsorption physics as PAI but carried to completion, including the occasional large cluster from a δ -electron. It is the accepted standard for gaseous-detector cluster statistics. It knows nothing about our PCB or mylar; it only does gas. That is why the pipeline exports each track's gas-crossing segment (entry point, direction, momentum) from Geant4 and lets Heed *re-ionize* that segment when we need full fidelity — the division of labor recommended by the Geant4-Garfield++ interface paper: Geant4 for everything solid, Heed for ionization in the gas.

Magboltz — billions of collisions to get four numbers

Magboltz (Biagi) answers a different question: not "what does one fast particle do", but "how does a population of *slow* electrons behave in this gas under this field?" It takes a library of measured electron-molecule cross sections (elastic, all the excitation levels, ionization, attachment — maintained for decades and continuously refined against swarm experiments) and literally simulates electrons bouncing collision-by-collision in a uniform field — billions of collisions — until the statistics converge. Out come the transport coefficients we need at each field value: **drift velocity, longitudinal and transverse diffusion, Townsend multiplication coefficient, attachment**, plus the rate of every individual excitation level. We run it once per gas over a log grid of fields covering both the drift (~ 100 V/cm– 1 kV/cm) and amplification (~ 30 – 60 kV/cm) regimes — order an hour per gas — and cache the resulting table forever.

The excitation-level bookkeeping is what makes **Penning transfer** possible: in an Ar or Ne mixture many excited noble-gas states carry more energy than the quencher's ionization potential, so a fraction r of those excitations ionize a quencher molecule instead of being lost. That fraction can't be computed from first principles — it is fitted to measured gain curves, and published values exist only for some binary mixtures (Ar/Iso 10 % $\rightarrow r = 0.40$, Ne/CO₂ 90:10 $\rightarrow 0.577\dots$). Ignoring it makes simulated Micromegas gains wrong by large factors, which is why we treat r as a first-class, bracketed systematic for the Ne mixes where no measurement exists.

Garfield++ — the umbrella, and the avalanche question

Garfield++ is the framework that ties the specialists together: it hosts Heed (as `TrackHeed`), reads Magboltz tables (as `MediumMagboltz`), holds the field description, and transports drift electrons. For avalanches it offers a ladder of fidelity: `AvalancheMicroscopic` follows *every electron*

collision-by-collision through the 150 μm gap — the gold standard for gain distributions and mesh transparency, but its cost scales with avalanche size (seconds per avalanche at gain 10^4 — unusable as a per-event workhorse); **AvalancheMC** is a faster, coarser Monte Carlo; and the **parameterized level** — a Polya-distributed gain with mean $\bar{g}$ and shape θ , times a mesh transparency factor — costs nothing per event. Our tiering: run **AvalancheMicroscopic** *once* per (gas, voltage) point on a few thousand single electrons to extract $\bar{g}$, θ and transparency, freeze those numbers into the fast Stage-B sampler, and anchor the absolute gain scale to measured curves. This is exactly the pattern ATLAS NSW digitization and the CAST/IAXO chains use.

Degrad — held in reserve

Degrad (also Biagi) is the microscope for one specific corner: what happens in the few nanoseconds after a photon photo-absorbs *in the gas* — shell-resolved absorption, Auger cascades, shake-off —

giving the electron-count distribution to better than 3 % below 20 keV. We only need that if escape-peak topology or Fano-limited resolution becomes decision-relevant, so it is a Phase-4 validation item, not part of the campaign loop.

Why not one big coupled program?

Garfield++ ships an official Geant4 interface that hands tracks over in-flight inside one binary. It is the right tool for a one-off validation run, and we plan to use it exactly that way. But for a 500-run campaign it has a structural problem: the slowest stage (avalanching) throttles everything, and changing a threshold or a voltage re-runs the particle transport too. The staged pipeline inverts that: the expensive Geant4 pass runs once per beam configuration, and the cheap response stages re-run in seconds per (gain, threshold, gas) variation. Even the interface paper's own tutorial warns the coupled route "will hardly be fast".

8 · The VMM3a front-end, explained simply

Neither of us lives in electronics land, so this section builds the picture from zero. It summarizes the research pass we just finished on how to model the readout chip honestly — what the chip does to our signal, which published models exist, and what we decided to implement.

What the chip is

The VMM3a is the custom chip (an ASIC, designed for the ATLAS New Small Wheel) that reads the pads. One chip handles **64 channels**; one pad = one channel, so a 1280-pad wedge needs 20 of them. There is no trigger: every channel that sees a signal above its threshold reports itself, timestamped, into the streaming DAQ — which is why the photon background question and the bandwidth question are really the same question.

The journey of one signal

Pad → charge amplifier

collects the avalanche charge onto a capacitor → a voltage step



Shaper

turns the step into a smooth bump of chosen width (peaking time 25 / 50 / 100 / 200 ns) — filters out noise



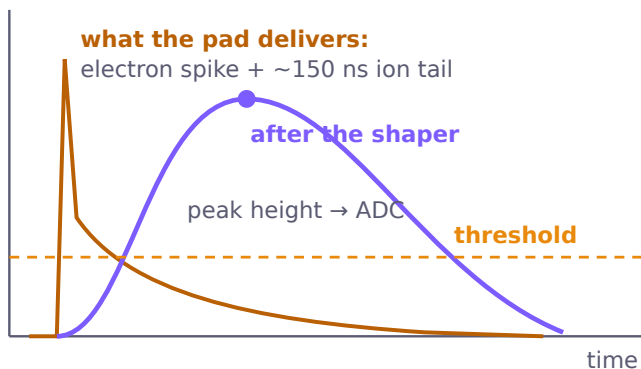
Discriminator + peak ADC + timing

bump taller than threshold? → record its height (10-bit ADC, ~8-bit effective) and its time (sub-2.5 ns) — otherwise stay silent



Neighbor logic

a firing channel forces its neighbors to report too, even below threshold



The physical signal from a pad is a very fast spike (the avalanche electrons arriving,

~ns) followed by a slow tail lasting ~150 ns (the ions drifting back across the amplification gap — most of the charge is actually in the tail). The **charge amplifier** integrates all of that onto a capacitor. The **shaper** then deliberately smears the result into a smooth bump whose height is proportional to the collected charge; you choose the bump width ("peaking time"). Why smear a perfectly good signal? Noise: a narrow measurement is sensitive to fast noise, a wide one averages it away — but too wide and pulses start piling on each other at high rate. At the 200 ns setting the bump is wider than the ion tail, so essentially *all* the charge contributes to the height; at faster settings part of the tail is cut off (the "ballistic deficit") and the bump under-reports the charge by a known factor.

The three numbers that rule everything

Gain — how many millivolts of bump height per femtocoulomb of charge (settable, 0.5–16 mV/fC). **Threshold** — the bump height below which the channel stays silent. And

noise (ENC) — the channel's baseline jitters as if a small random charge were arriving continuously; it is quoted as an "equivalent noise charge" in electrons. The key fact for us: **ENC grows with the capacitance of whatever is connected to the input**, and our pads are large — 75–170 pF — so the noise floor is several thousand electrons. That sets how low the threshold can usefully go (a few fC), and it is why the threshold question and the gas-gain question are coupled: the gas gain must lift a handful of primary electrons safely above a noise floor fixed by pad geometry. For calibration: the expected signal is ~ 21 fC, and the chip measures 62 fC–2 pC over its range.

Two subtleties that directly affect our physics conclusions

Neighbor logic. Charge from one track usually spreads over 2–3 pads; the outer pads often fall below threshold and would be lost — biasing cluster size and total charge low. The VMM has a fix in hardware: when a channel fires, its neighbors are read

out too, threshold or not (even across chip boundaries). We must model this, because our electron-vs-photon discrimination partly rests on cluster *topology* (a MIP smears charge along the track; a 6–8 keV photoabsorption is one fat point on 1–2 pads), and neighbor logic changes exactly that observable.

ADC saturation. The peak ADC has ~ 8 effective bits. At high gas gain, a point-like fluorescence photoabsorption (~ 250 electrons before gain) can rail the ADC while MIP pads stay comfortably in range — which *compresses* the very charge difference the discrimination cut uses. So saturation is not a detail; it is part of the ROC curve, and we simulate it explicitly.

What we just decided, and how it plugs in

The research pass found two published ways to model the chip: the ATLAS Athena `VMM_Shaper` — the exact closed-form shaper response $h(t)$, about 20 lines of code, including peak detection, ballistic-deficit factor and neighbor-logic emulation — and

a simpler semi-Gaussian proxy used by the ESS/CERN groups. Our decision, in plain terms: **for everything the campaign needs, we don't simulate waveforms at all.** At 200 ns peaking, bump height $\approx$ collected charge, so Stage C takes the per-pad charges from Stage B and applies: (1) Gaussian noise at the ENC for our pad capacitance, (2) a threshold scan in fC (also quoted in primary-electron equivalents, so the conclusion survives gain-scale uncertainty), (3) neighbor logic, (4) ADC quantization + saturation. That produces exactly what the DAQ would record — which pads report, with what ADC values — and from there the efficiency, fake-hit and ROC numbers. The full Athena $h(t)$ waveform model is kept in the back pocket (Phase 4) for questions that genuinely live in the time domain: hit timing, time-walk, and pile-up in the trigger-less stream. None of the Phase 1–3 deliverables need it.

VMM questions for you (whenever convenient — you have the hardware experience with these boards):

1. Which peaking time and gain setting does BASKET currently run (prototype and planned)? We assumed 200 ns and mid-range gain.
2. Do you have a measured noise (ENC or threshold-in-fC) figure from the prototype at real pad capacitance? One number would anchor Stage C's noise model to reality.
3. Is neighbor logic enabled in your DAQ configuration?

Bon voyage — see you on the other side. Questions in the purple boxes, everything else is context. — D.